Disaggregation Plug Load Management Applications

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Project Overview

With the world's increasing dependence on technology, growing environmental concerns over energy sourcing in peaker plants (**Figure 1**), and the roll-out of Time of Use energy billing rates in California, it is imperative to minimize grid loads during peak demand hours and to minimize overhead costs to residents and businesses. This project aims to develop automation applications for various residential and industrial settings that allow users to audit and manage energy usage around a home/facility then make informed decisions on certain types of devices/appliances. By simulating a home environment, the authors test and refine the effectiveness of such system on real world scenarios.

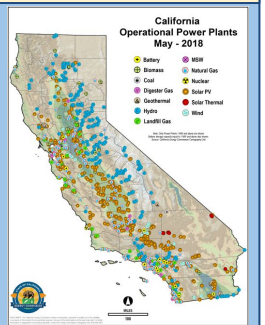

Figure 1: Concentration of energy sources in CA. Gases (pink, green, purple) are prevalent in populated areas (CEC).

Research Objectives

Figure 2: Smart Meters are installed at each house in California by major utilities

- Present the energy usage data and actionable information provided by energy management authorities using IoT to empower users to save energy and minimize cost.
- Augment and extend contextual information using additional smart-home sensors.
- Provide multiple modalities of access to target different demographics while maintaining universality, resilience, and independence among applications.

Developed Solutions

These solutions were created to control devices and to audit, manage, and present energy usage

- PlugLoad Perceptoscope (iOS)
  - Spatial object tracking and object recognition via neural networks
  - Present energy data through AR graphs
  - Use analyzed summary data to suggest focus points and provide energy-saving tips
- CalPlug Energy Auditor (Google Asst., Alexa)
  - Compatible with Android devices, Home speakers, Alexa enabled devices
  - Uses natural language processing to determine requested actions
  - Retrieves recent usage data to estimate energy bill for recommend savings options
- CalPlug Energy Stats (Web)
  - Displays both individual and aggregate energy data graphically in responsive UI.
  - Provides bill and usage estimation to offer possible efficient device sub. from provider
- Audit Bot (Slack, Discord)
  - Targets Slack orgs. and Discord servers
  - Follows common bot interface paradigms
  - Notifies members of high energy demand
  - Sends rebate options to user based on energy usage patterns of the user

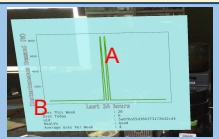

Figure 3: Perceptoscope showing microwave usage data. Shows graph of recent instantaneous demand readings (A) and summarized usage data (B).

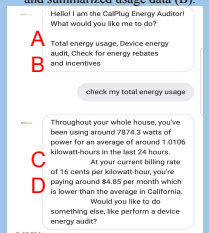

Figure 4: CPEA conversation, Initial prompt (A), command to check total usage (B), usage data (C), and cost estimate(D).

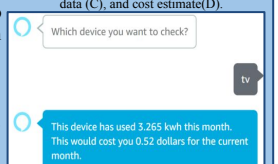

Figure 5: Alexa conversation illustrating individual device auditing.

Acknowledgements

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Developed Solutions (cont'd)

Figures 5,6,7: Screenshots of Slack (5), Discord (6), and Web UI (7) demonstrating auditing and notifying capabilities. Using interfaces appropriate for medium, they independently present relevant information to user. Web UI allows for time-based disag. comparison

System Accomplishments

- Separation of concerns & abstraction between front and back ends
- Focus on reasonable & actionable design decisions for medium
- App resilience to backend changes under app-layer protocol

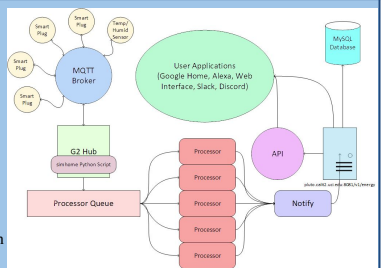

Figure 8: Architecture diagram for energy auditing systems

Conclusion & Future Developments

These applications were developed successfully and are flexible to changes in backend structure, provided the application-layer protocol is maintained. Future developments include:

- Management of environment-based device automation.
- Implementation of user profile handling
- Further use of common IoT and data security measures

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